National University Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

CL-1004
Object Oriented Programming
Assignment 02

Objectives:
DMA Recursion Structure

Total Questions: 08	**Total Marks: 110**

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at the start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function and about its functionality.
3. Mention comments where necessary such as comments with variables, loop, classes etc. to increase code understandability.
4. Use understandable names of variables.
5. Proper indentation of code is essential.
6. Write a code in C++ language.
7. Make a Microsoft Word file and paste all your C++ code with all possible screenshots of every task **output in Microsoft Word and submit the word file. Do not submit .cpp file.**
8. First think about statement problems and then write/draw your logic on copy. 9. After copy pencil work, code the problem statement on MS Studio C++ compiler. 10. At the end when you have done your tasks, attached C++ created files in MS word file and make your submission on Google Classroom. (Make sure your submission is completed).
9. Please submit your file in this format **22F_XXXX_A1_Sec.zip**.
10. Do not submit your assignment after the deadline. Late submission is not accepted.
11. Do not copy code from any source otherwise you will be penalized with negative marks or zero.

Part 1: | DMA	10*2=20

1- Imagine you are working on a scientific research project that involves collecting data from various sensors placed at different locations. Each sensor generates a varying number of data points over time, and you need to organize this data efficiently for analysis. To do this, you decide to create a program that builds a matrix to store the sensor data, where each row corresponds to a different sensor, and the number of columns varies for each sensor based on the data points collected.
Design a C++ program to build a matrix with a different number of elements in each row (different number of columns in each row) using a two-dimensional dynamic array.

For Example:

```
Matrix is
1  2  3
4  5  6  7  8
1  2
```

Your program must contain two functions. One for filling the elements into your two-dimensional array and the other for printing that array or matrix.

2- Input a sentence from the user. Use full stop, space and comma as word separators. Each word should be stored in a 2D array whose columns vary in size and each row stores one word as a NULL terminated string. For example, if the user inputs:
Hello, how are you?
It should be stored as:

H	e	l	l	o	NULL
h	o	w	NULL		
a	r	e	NULL		
y	o	u	?	NULL	

Part 2: | Recursion	**10*4=40**

1. Define a recursive function named "int power (int b, int e)" to calculate the power of two numbers.
2. Define a recursive function "str reverse (string str)" to reverse the given string.
3. Implement a recursive function to check if a given string is a palindrome (reads the same forwards and backwards).
4. (Visualizing Recursion) It's interesting to watch recursion "in action." Modify the factorial function to print its local variable and recursive call parameter. For each recursive call, display the outputs on a separate line and add a level of indentation. Do your utmost to make the outputs clear, interesting and meaningful. Your goal here is to design and implement an output format that helps a person understand recursion better. You may want to add such display capabilities to the many other recursion examples and exercises throughout the text.

Part 3: | Structure	**20*1=20**

Problem 07:

Write the code in C++ using Structure concept to develop a basic scientific calculator with following set of functionalities.

- To calculate the inverse of the value.
- Squares the value.
- Cubes the value.

- Calculates exponential values.
- Calculates the square root of the value.
- Calculates the cube root of the value.
- Calculates powers based on the constant e (2.718281828)
- Calculates the sin, cos, tan, cosec, sec and cot.

However, all crude operations like addition, multiplication etc. will be part of the calculator.

Part 4: | Struct & DMA	**30*1=30**

Problem 08:

Make a simple Airline Seat Reservation System. The system should allow passengers to reserve seats on a flight. Each flight has a fixed number of rows and columns of seats, and the system should use dynamic memory allocation to manage seat reservations efficiently.
Design a program for the Airline Seat Reservation System using structures and dynamic memory allocation with the following requirements:

1. Define a structure named Seat with members to represent the seat's row, column, and reservation status (e.g., reserved or available).
2. Create a function named initializeFlight that takes the number of rows and columns as parameters and initializes the flight's seating arrangement, dynamically allocating memory for the seats.
3. Create a function named reserveSeat that takes a flight and seat row and column as parameters. It should reserve the seat if it's available (i.e., not already reserved) and return a confirmation message.

4. Create a function named cancelReservation that takes a flight and seat row and column as parameters. It should cancel the reservation if the seat is currently reserved and return a confirmation message.
5. Create a function named displaySeatingArrangement that displays the current seating arrangement for a flight, showing 'R' for reserved seats and 'A' for available seats.

In the main function, create a flight with a specific number of rows and columns, allow passengers to reserve and cancel seats, and display the seating arrangement after each operation.